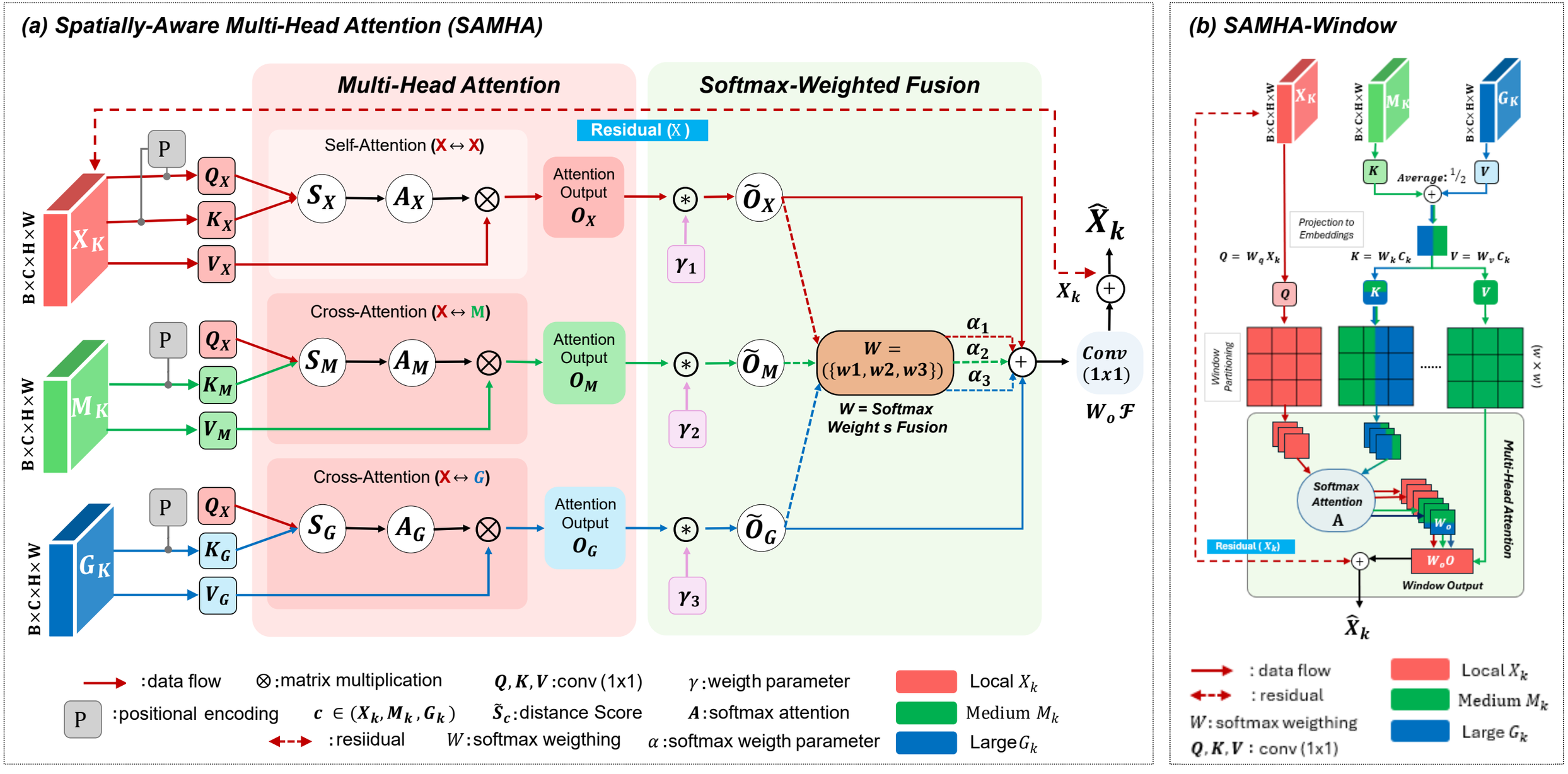
(a) Spatially-Aware Multi-Head Attention (SAMHA)
Multi-Head Attention
Softmax-Weighted Fusion
Residual (X)
Self-Attention (X ↔ X)
Cross-Attention (X ↔ M)
Cross-Attention (X ↔ G)
Attention Output O_X
Attention Output O_M
Attention Output O_G
W = Softmax Weight s Fusion
Conv (1x1)
$W_o\mathcal{F}$
$\hat{X}_k$
X_k
B×C×H×W
:data flow
⊗:matrix multiplication
Q, K, V:conv (1x1)
γ:weigth parameter
:positional encoding
$c \in (X_k, M_k, G_k)$
$\tilde{S}_c$:distance Score
A:softmax attention
:resiidual
W:softmax weigthing
α:softmax weigth parameter
Local X_k
Medium M_k
Large G_k
(b) SAMHA-Window
Average: 1/2
Projection to Embeddings
$Q = W_q X_k$
$K = W_k C_k$
$V = W_v C_k$
Window Partitioning
Softmax Attention A
Multi-Head Attention
Residual (X_k)
$W_o O$
Window Output
: data flow
: residual
W: softmax weigthing
Q, K, V : conv (1x1)
Local X_k
Medium M_k
Large G_k